FreshCart

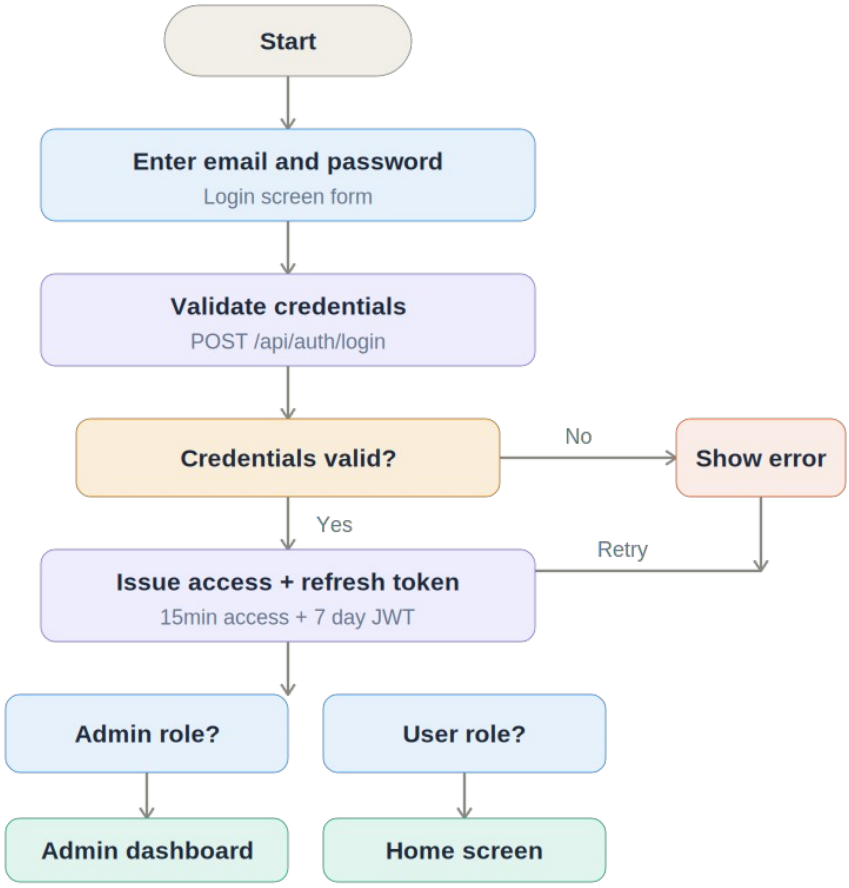
Activity Diagrams

Technical Documentation — Application Workflows

This document presents the key activity diagrams for the FreshCart mobile application, illustrating the main user workflows and business processes including authentication, product scanning, checkout, payment, and administration.

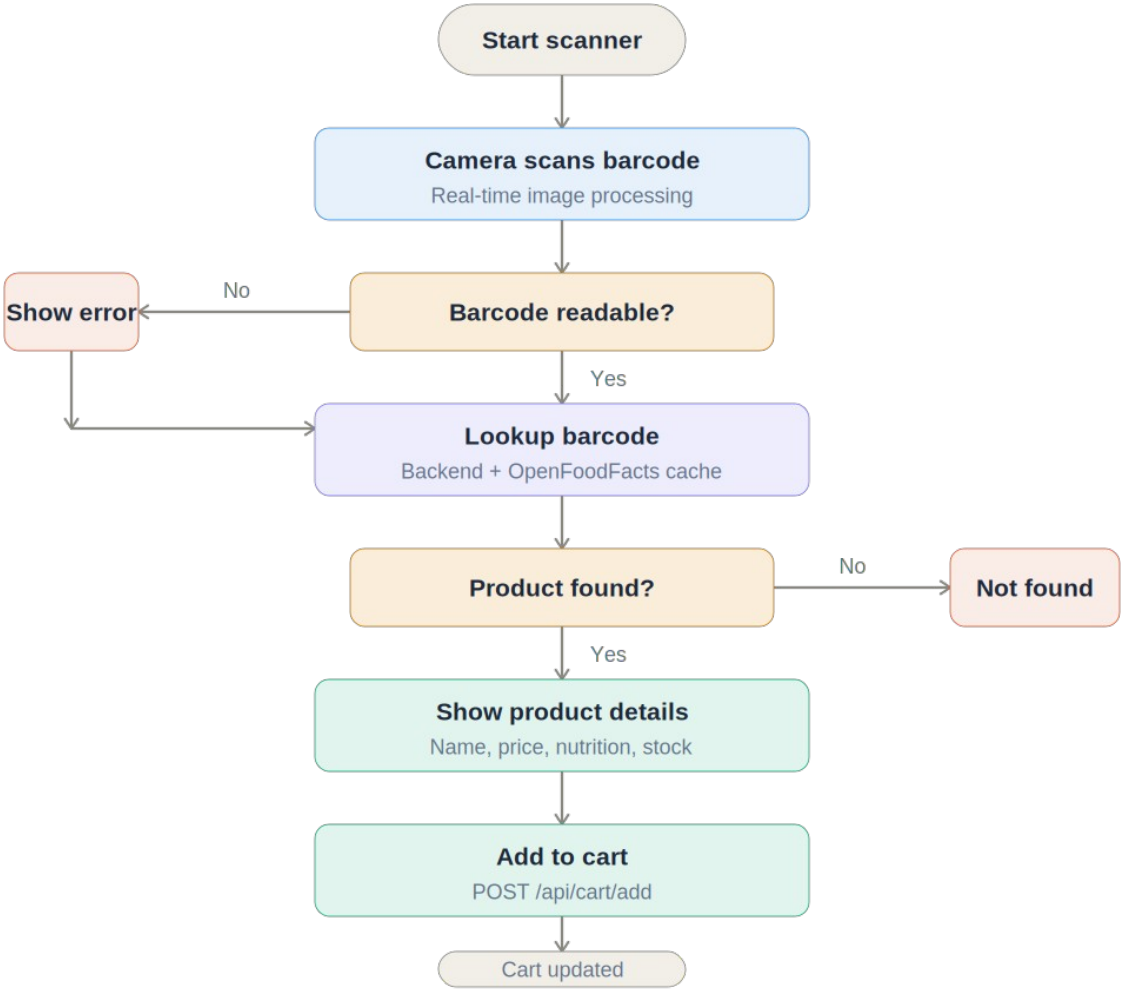
1. User Login Flow

Illustrates the authentication process from entering credentials to role-based redirection. Includes error handling and JWT token issuance with a 15-minute access token and 7-day refresh token.



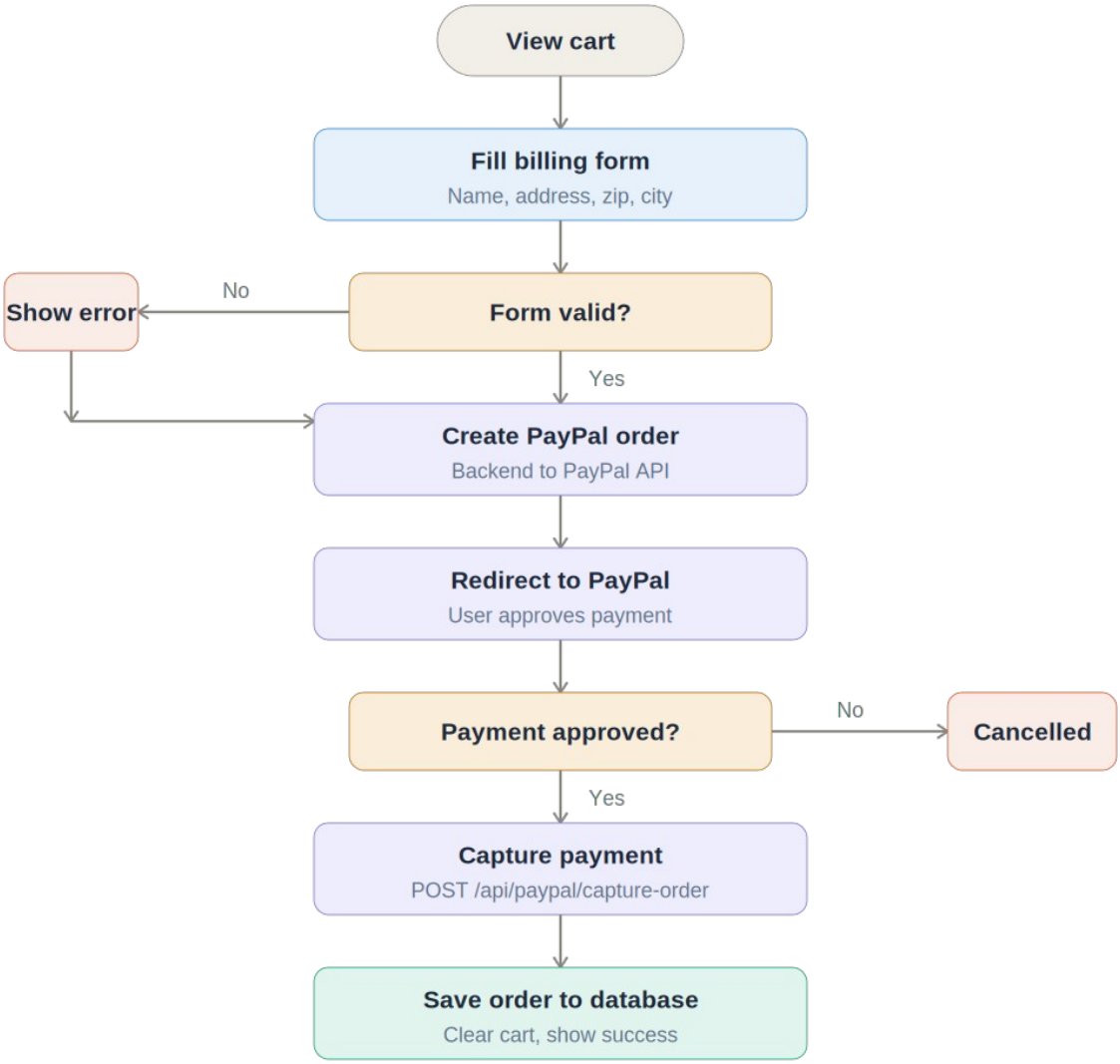
2. Barcode Scan to Cart Flow

Shows the complete barcode scanning workflow from camera activation to adding a product to the cart. Includes OpenFoodFacts lookup with in-memory caching and error handling for unreadable or not-found barcodes.



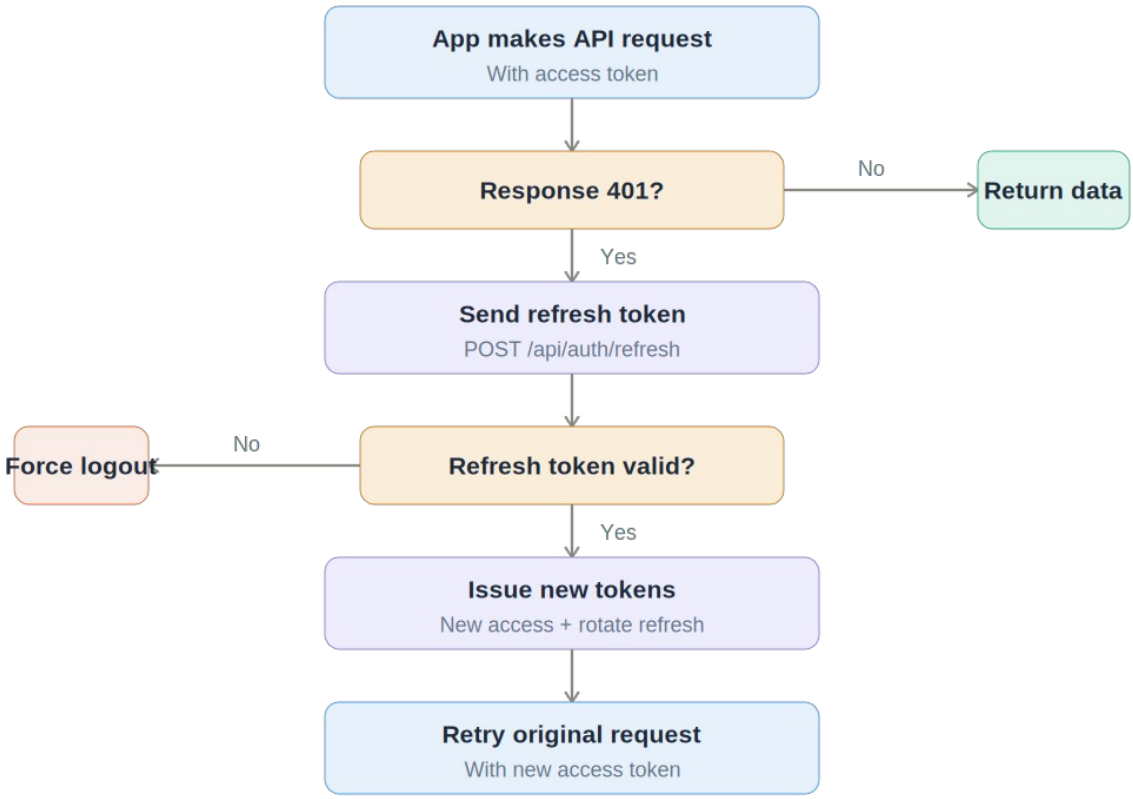
3. Checkout and PayPal Payment Flow

Covers the full checkout process from cart review to order confirmation. Includes billing form validation, PayPal order creation, user approval redirect, payment capture, and order storage in the database.



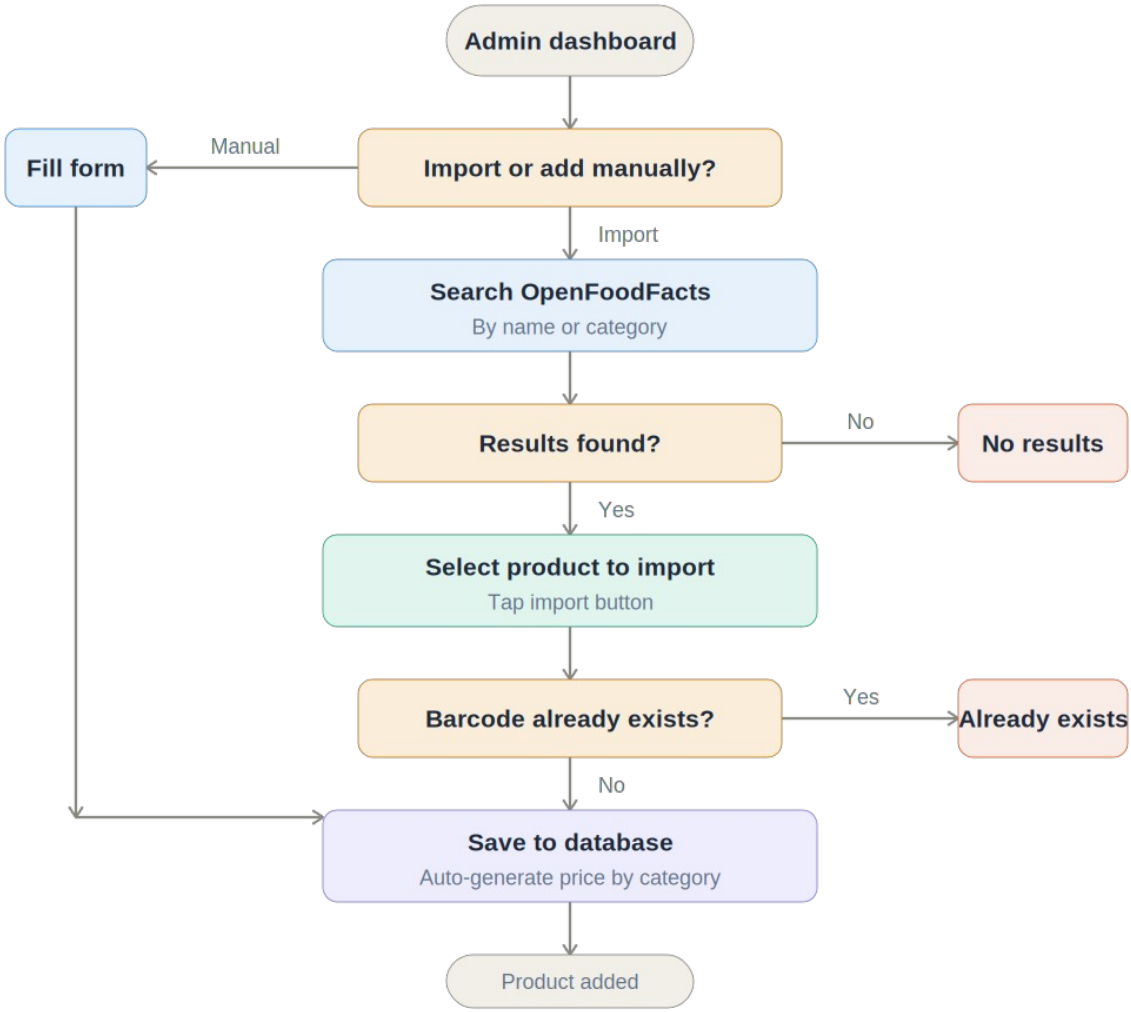
4. Token Refresh Flow

Describes the automatic token refresh mechanism. When a 401 response is received, the app transparently exchanges the refresh token for new tokens and retries the original request. Forces logout if the refresh token is invalid or expired.



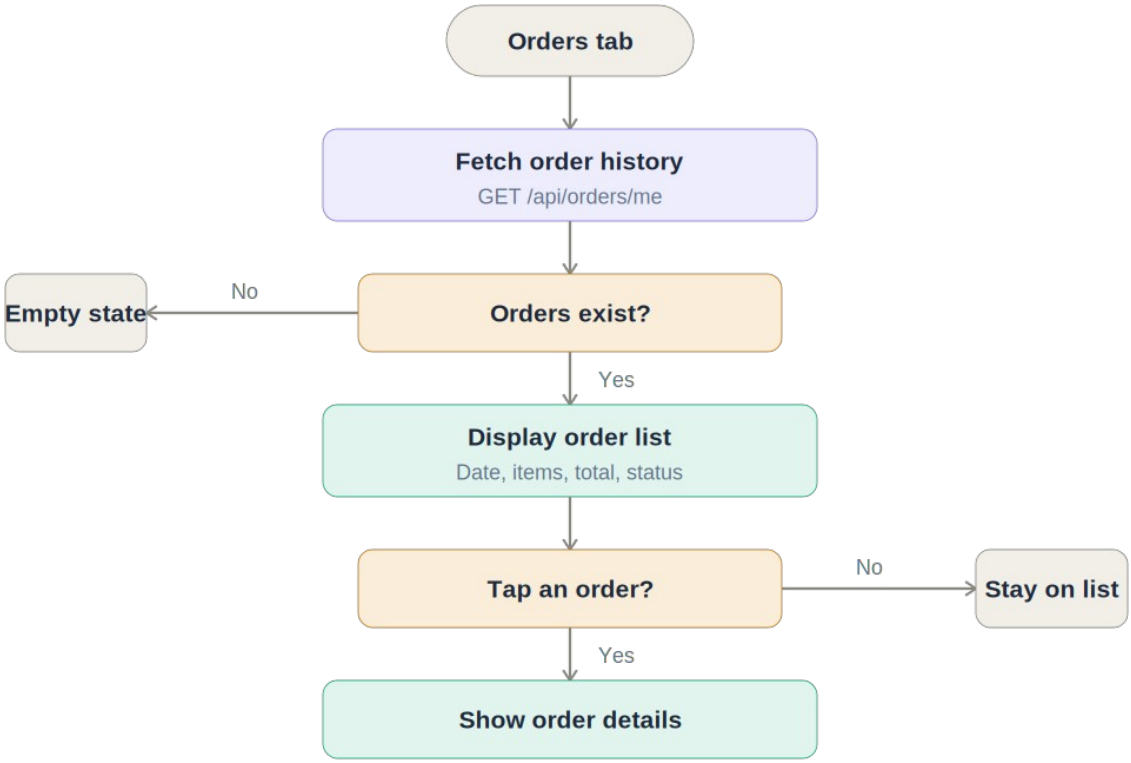
5. Admin Product Import Flow

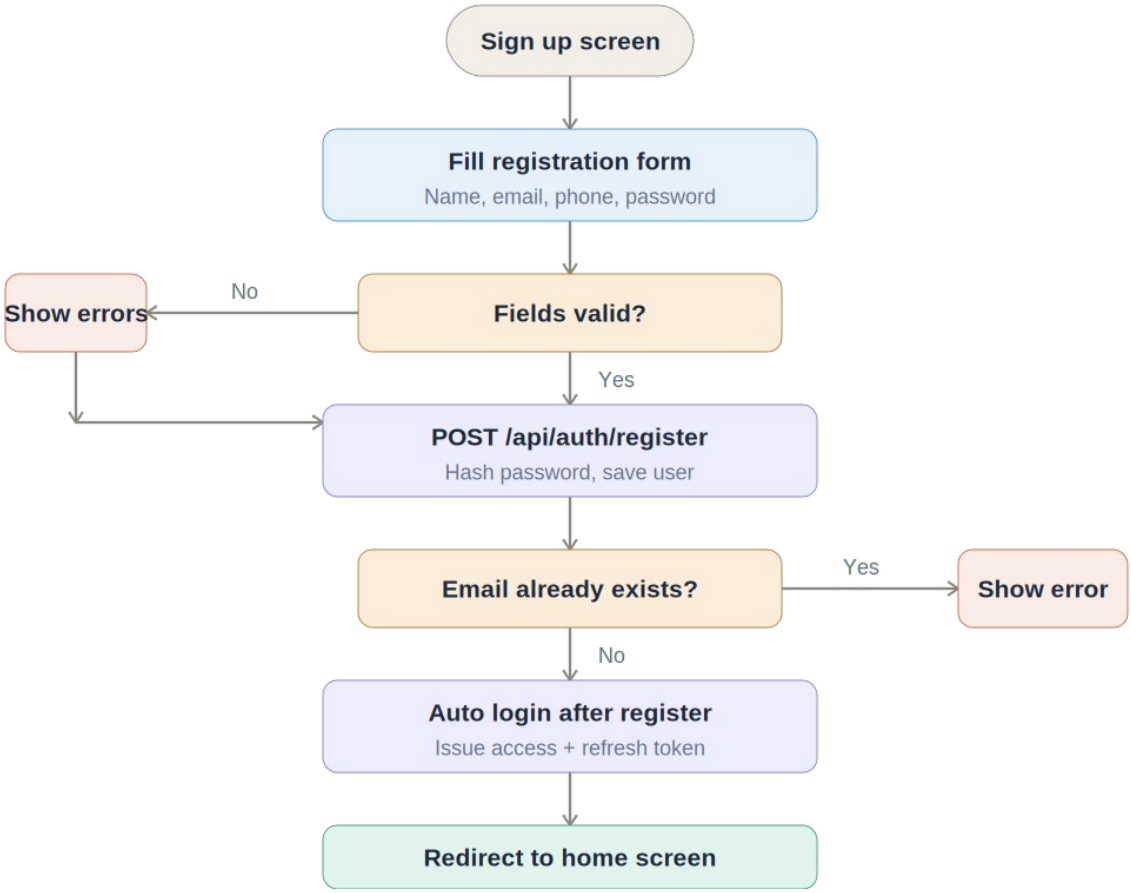
Details how an admin can import products from OpenFoodFacts by name or category. Includes duplicate barcode detection and automatic price generation based on product category.



6. Order History Flow

Shows how a user accesses and browses their purchase history, from fetching orders via the API to viewing individual order details.



7. User Registration Flow

Covers the registration process including form validation, duplicate email detection, password hashing, and automatic login after successful registration.